

# Inert Show-Cue Simulator

Assess every channel, review every cue, replay every decision

*A composed teaching simulator with no external action path*

|                      |                                                  |               |                           |
|----------------------|--------------------------------------------------|---------------|---------------------------|
| <b>Lesson</b>        | 030 — capstone project                           | <b>Time</b>   | 240–300 minutes           |
| <b>Prerequisites</b> | Lessons 028 and 029                              | <b>Board</b>  | Arduino Mega 2560         |
| <b>Evidence</b>      | TP28, TP29, cue LEDs, RGB state, audit           | <b>Status</b> | host verified; bench open |
| <b>Energy class</b>  | E1: USB, passive switches, resistor-limited LEDs |               |                           |

**Boundary.** This project composes synthetic switch observations with opaque visual cue labels. It has no relay, transistor-driven load, initiator, show connector, antenna, transmitter, callback, generic output sink, captured protocol, or external equipment interface. A lit cue is a teaching-state presentation only. USB power removal is the physical stop; software cancellation and shutdown are lifecycle behavior, not an emergency stop.

## 1 Mission and evidence chain

eight synthetic observations → channel assessment

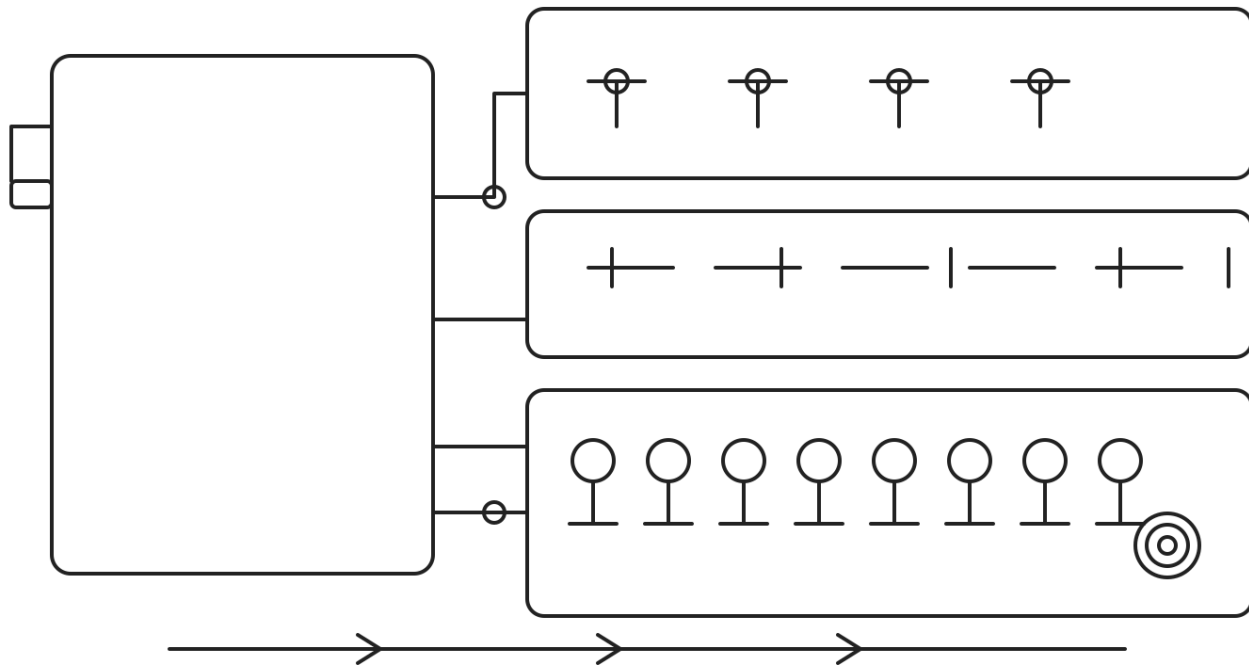
→ reviewed cue decision → inert LEDs + audit + digest.

Lesson 028 supplies bounded channel assessments. Lesson 029 supplies a copied plan, supplied-time scheduler, and append-only audit. Lesson 030 coordinates their lifecycles and decisions without owning an endpoint or performing I/O.

### 1.1 Learning outcomes

- compose two deterministic components through value-only contracts;
- map plan positions to channels without interpreting opaque cue IDs;
- explain Permit, Hold, and Fault evidence gates;
- apply total simultaneous-event precedence;
- reconcile snapshots, audit records, and a deterministic trace digest;
- distinguish active safe-state evidence from resource-release evidence.

## 2 Orientation and unpowered inspection



*Alternative text: pencil orientation plate showing one USB-powered Mega, eight synthetic input switches, five pull-up operator buttons, eight separately resistor-limited cue LEDs, a separately resistor-limited common-cathode RGB state LED, TP28 at the selected synthetic input, and TP29 at the selected cue output. No external-load connection is present.*

| Role              | Mega pin                 | Connection                                               |
|-------------------|--------------------------|----------------------------------------------------------|
| cue indicators    | D22–D29                  | pin to 1 k $\Omega$ , LED anode; cathode to GND          |
| operator controls | D30–D34                  | Review, Run, Confirm, Skip, Cancel to GND; pull-up       |
| synthetic editor  | D35–D38                  | passive switch inputs used to choose test states         |
| RGB state         | D5–D7                    | red, green, blue through separate 1 k $\Omega$ resistors |
| Lesson 030 TP28   | selected synthetic input | logic-level test point to GND                            |
| Lesson 030 TP29   | selected cue output      | Mega-pin side of resistor to GND                         |

**Unpowered prediction.** Verify LED polarity, one resistor per LED channel, switch-to-ground paths, common ground, and no connection outside the USB-powered fixture. Record resistor values: \_\_\_\_\_.

## 3 Current, pins, and ownership

At most one cue channel and one RGB channel are commanded. A conservative zero-forward-voltage bound gives

$$I_{\text{channel}} \leq \frac{5.0 \text{ V}}{1000 \Omega} = 5.0 \text{ mA}, \quad I_{\text{LED, total}} \leq 10.0 \text{ mA}.$$

This is a design calculation, not a measured result.

| Quantity              | Predicted     | Measured | Calculated        | Interpretation |
|-----------------------|---------------|----------|-------------------|----------------|
| USB/5 V rail          | at most 5.0 V | _____    | —                 | _____          |
| selected cue resistor | 1 k $\Omega$  | _____    | $I = (V - V_f)/R$ | _____          |
| active RGB resistor   | 1 k $\Omega$  | _____    | $I = (V - V_f)/R$ | _____          |
| worst LED load        | at most 10 mA | _____    | _____             | _____          |

Construction is inert. Initialization starts the assessor, then the scheduler and its audit session; failure rolls back in reverse order. Shutdown first forces the scheduler's no-visible-cue state and Shutdown record, then stops the assessor. The external audit storage deliberately remains readable.

## 4 Plan-to-channel composition

Cue values remain opaque visual labels. The separate map is indexed by plan position and is copied during construction.

| Plan index | Cue label | Offset   | Visible for | Assessed channel |
|------------|-----------|----------|-------------|------------------|
| 0          | 3         | 2000 ms  | 2000 ms     | 5                |
| 1          | 7         | 7000 ms  | 2000 ms     | 2                |
| 2          | 12        | 12000 ms | 2000 ms     | 7                |
| 3          | 29        | 17000 ms | 2000 ms     | 0                |

**Do not infer a channel from a cue.** Cue 3 maps to channel 5, cue 7 maps to channel 2, cue 12 maps to channel 7, and cue 29 maps to channel 0. The cue LED is addressed with the scheduler's bounded `cueIndex`; the selected assessment uses the explicit map.

The simulator borrows the assessor, scheduler, and audit. Those objects and their caller-owned storage must outlive it. It allocates no heap storage and retains no observation pointer.

## 5 Complete-frame transaction

Every accepted update contains exactly eight unique channel observations. Order does not matter: the simulator canonicalizes them by channel before identity comparison and digest calculation. Null storage, a wrong count, duplicate or missing channels, invalid values, and ambiguous time are rejected.

```
const adk::InertShowInput input = {
    observations, 8, operatorInput
};
const adk::Status status = simulator.update(now, input);
const adk::InertShowSnapshot view = simulator.snapshot();
```

The pointed-to observations are consumed synchronously; no pointer survives `update()`. An identical complete frame repeated at the same timestamp is idempotent. A changed frame at that timestamp is invalid. The example samples and decides once per distinct `millis()` value.

## 5.1 Evidence gate

| Assessment     | Gate   | Result                                            |
|----------------|--------|---------------------------------------------------|
| Closed         | Permit | confirmation may make the reviewed cue visible    |
| Open           | Hold   | cue stays hidden; wait for a fresh complete frame |
| ShortSimulated | Hold   | cue stays hidden with a distinct condition        |
| Stale          | Hold   | cue stays hidden until fresh evidence arrives     |
| Unavailable    | Hold   | cue stays hidden; no automatic bypass             |
| Contradictory  | Fault  | latch fault and guarantee no visible cue          |

Recovery from a nonterminal Hold never resumes a cue automatically. The operator returns through review and supplies a fresh confirmation.

## 6 State and precedence

| Show state | Meaning                                           | Circuit-native evidence                  |
|------------|---------------------------------------------------|------------------------------------------|
| Startup    | coordinator not active                            | all cue LEDs inactive                    |
| Review     | operator review gate held                         | review RGB pattern; cues inactive        |
| Ready      | waiting or confirming                             | pending-state RGB pattern; cues inactive |
| Running    | scheduler has a visible cue                       | exactly one cue LED and TP29             |
| Held       | review or evidence interrupted progress           | held RGB; cues inactive                  |
| Complete   | plan ended                                        | completion RGB; cues inactive            |
| Cancelled  | cancellation latched                              | cancelled RGB; cues inactive             |
| Fault      | contradiction, input, invariant, or audit failure | fault RGB; cues inactive                 |

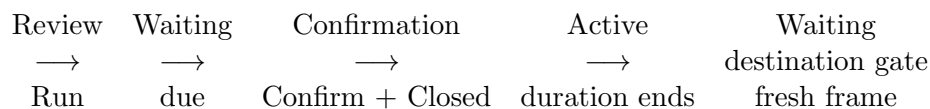
For a newly accepted full frame the total order is:

Cancel  $\succ$  invalid chord  $\succ$  evidence Fault/Hold  $\succ$  Review release  $\succ$  phase action  $\succ$  time

Frame identity is checked first. Cancel therefore wins over simultaneous Confirm, Skip, or Run, but it does not turn a changed same-timestamp frame into a second accepted transaction. More than one non-cancel edge is an invalid chord. Review release prevents confirmation and immediately removes a visible cue.

## 7 Timing and no-catch-up rule

Run fixes the plan epoch; late decisions never move cue offsets. Confirmation is accepted only inside its inclusive window. A delayed update records and hides elapsed work, but does not briefly activate every missed cue. Advancing from one cue reaches a hidden waiting state first; a later fresh frame assesses the destination channel before requesting confirmation.



At TP29 predict inactivity before accepted confirmation, a bounded active interval after it, and immediate inactivity on Hold, Cancel, Fault, completion, or shutdown. A lit LED proves only the presentation path.

## 8 Audit, replay, and digest

The scheduler remains the sole writer to `CueAuditBuffer`; the coordinator does not duplicate records. The ledger never overwrites old evidence and reserves room for Shutdown before a run begins. If a required transition cannot be recorded, no corresponding visible transition occurs.

```
make inert-show-test
make inert-show-replay TRACE=tests/fixtures/inert_show_happy.trace
make inert-show-audit TRACE=tests/fixtures/inert_show_happy.trace
```

The versioned trace contains supplied ticks, eight synthetic observation tokens, and debounced operator values. Replay emits stable snapshots; audit mode emits the stable Lesson 029 audit grammar. The 32-bit trace digest detects accidental replay differences. It is neither authentication nor a security control.

### 8.1 Trace prediction exercise

| Frame               | Prediction | Observed replay | Reconciliation |
|---------------------|------------|-----------------|----------------|
| Review + all Closed | _____      | _____           | _____          |
| Run at first epoch  | _____      | _____           | _____          |
| Confirm first cue   | _____      | _____           | _____          |
| mapped channel Open | _____      | _____           | _____          |
| Cancel + Confirm    | _____      | _____           | _____          |

For one transition, copy the snapshot sequence \_\_\_\_\_, audit sequence \_\_\_\_\_, and final digest \_\_\_\_\_. Explain any disagreement: \_\_\_\_\_.

## 9 Narrative Mega example

```
void loop ()
{
    const adk::TimePoint now (millis ());

    readSimulatorInput (now);
    updateSimulator    (now);
    showInertCues      ();
    showSimulatorState (now);
}
```

Objects appear in dependency order: runtime and claims, buttons, cue and state LEDs, synthetic source and assessor, plan and audit storage, scheduler, then simulator. Setup reads as acquire, configure, start. The loop reads as observe, decide, actuate. Presentation consumes one snapshot and cannot call back into the simulator. Serial replay is optional supporting evidence, never the only observation path.

## 10 Predict, observe, interpret

1. Before power, predict TP28 and TP29 outcomes for channel 5 Closed.
2. Start the fixture. Observe the acquisition sweep separately from the later all-cues-inactive safe state.
3. Hold Review and press Run. Predict TP29 remains inactive before confirmation even when the mapped channel is Closed.

4. Confirm cue 3. Compare the channel-5 TP28 state, selected assessment, cue-index-0 LED, TP29, and ordered audit entry.
5. Make the next mapped channel Open. Predict Hold and no cue; restore Closed, renew review, and confirm again.
6. Inject Contradictory. Predict latched Fault, inactive TP29, and no restart until reinitialization.
7. Invoke shutdown and record darkness separately. Only then remove USB.

## 11 Troubleshooting tree

| Observation                 | Stop, inspect, and interpret                         |
|-----------------------------|------------------------------------------------------|
| more than one cue LED       | remove USB; inspect bounded index presentation       |
| wrong cue for TP28 channel  | inspect the plan-position map; never use cue ID      |
| cue active while Open/Stale | remove USB; inspect gate and fresh-frame logic       |
| cue appears before Confirm  | remove USB; reject the fixture as valid evidence     |
| audit sequence stops        | expect AuditFull and no visible transition           |
| replay digest differs       | compare exact input frames, order, and tool identity |
| out-of-rail TP28 or TP29    | remove USB; inspect ground, switch, and resistor     |
| hot part or unstable USB    | remove USB immediately; troubleshoot unpowered       |

## 12 Claim–evidence–reasoning worksheet

| Claim                        | Evidence to record             | What it cannot prove |
|------------------------------|--------------------------------|----------------------|
| mapped cue is permitted      | TP28, assessment, snapshot     | external continuity  |
| presentation is correct      | cue identity and TP29 interval | resource release     |
| decision is replayable       | trace, audit bytes, digest     | authenticity         |
| shutdown requests inactivity | snapshot and inactive outputs  | power removal        |
| endpoint ownership released  | named release fixture or test  | emergency stop       |

Chosen claim: \_\_\_\_\_

Evidence: \_\_\_\_\_

Reasoning and limits: \_\_\_\_\_

## 13 Open hardware acceptance record

**Software gates do not constitute a physical result.** Keep this record open until the named board, circuit, instruments, measurements, date, and reviewer are present.

|                   |       |               |       |
|-------------------|-------|---------------|-------|
| Board/revision    | _____ | Commit        | _____ |
| LEDs and polarity | _____ | Resistors     | _____ |
| Supply/instrument | _____ | Tool versions | _____ |
| Operator/date     | _____ | Reviewer      | _____ |

- ☐ unpowered circuit, polarity, resistors, and prohibited external connections inspected;
- ☐ startup acquisition evidence predicted, observed, and interpreted separately from safe-state evidence;
- ☐ all eight synthetic states and explicit cue-channel mapping checked at TP28 and TP29;

- ☐ Permit, every Hold condition, contradiction, precedence, delayed update, cancel, and audit-full cases observed;
- ☐ replay snapshots, audit bytes, and digest reconciled;
- ☐ shutdown inactive state and resource release recorded as separate claims;
- ☐ USB power-removal result recorded.

Prediction: \_\_\_\_\_

Observation: \_\_\_\_\_

Interpretation and limits: \_\_\_\_\_

**Stop rule.** Remove USB power for a hot part, unstable supply, wrong polarity, missing resistor, unexpected indicator, out-of-rail TP28 or TP29, or schematic/breadboard disagreement. Do not attach an external load or real show-control connector. Software shutdown is never the physical stop.

## 14 Reproduce the software evidence

```
make inert-show-test
make host-test-exceptions
make inert-show-replay
make inert-show-audit
make arduino-Lesson030InertShowSimulator
make size-check
make lessons
make lessons-check
make site-check
make hardware-card LESSON=030 BOARD=mega2560
```

Record the exact AVR core, compiler, measured flash, and static RAM with the canonical sketch. Re-measure whenever source or toolchain identity changes. Do not substitute a budget or a host binary size for measured Mega evidence.

## 15 Reference trail

- ADK `inert_channel_assessor.h`, `inert_cue_scheduler.h`, `inert_show_simulator.h`, focused tests, replay fixture, and canonical Lesson 030 sketch;
- ADK development, testing, safety, packaging, and PDF policies;
- Lessons 028 and 029 for dependency contracts and evidence limits;
- Arduino Mega 2560 Rev3 datasheet for the exact board profile.

The searchable HTML reference is the primary accessible route. This printable PDF uses real text, headings, lists, and monochrome graphics but does not claim PDF/UA conformance.